

MOOSE JAW, SK, Canada

WMO: 715390

Lat: 50.332N

Lon: 105.537W

Elev: 1893

StdP: 13.72

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-18.4	-30.8	1.0	-23.3	-25.1	1.5	-18.3	32.2	25.4	29.6	21.8	8.4	280	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	89.2	65.4	85.4	64.4	81.8	63.1	70.0	81.4	67.6	79.4	65.6	77.2	11.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	104.5	74.6	63.7	94.7	72.7	61.4	87.2	70.5	35.2	81.5	33.2	79.1	31.5	77.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	24.9	22.2														
			DB		-31.2	95.9	5.6	4.0	-35.2	98.8	-38.5	101.1	-41.6	103.4	-45.7	106.3
			WB		-31.4	74.4	5.5	2.8	-35.4	76.4	-38.6	78.1	-41.7	79.7	-45.7	81.7

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	10.1	12.4	24.8	40.1	52.0	61.1	66.4	64.7	55.5	41.4	26.1	14.1	(6)
(7)		DBStd	23.06	15.88	14.87	14.56	10.33	8.62	6.56	5.49	6.39	8.32	9.29	12.64	14.92	(7)
(8)		HDD50	5635	1238	1052	782	319	79	3	0	0	39	293	717	1113	(8)
(9)		HDD65	9638	1703	1472	1246	747	410	151	49	84	300	732	1166	1578	(9)
(10)		CDD50	1694	0	0	1	22	141	336	508	456	203	26	1	0	(10)
(11)		CDD65	222	0	0	0	0	7	34	92	75	14	0	0	0	(11)
(12)		CDH74	3444	0	0	0	15	187	478	1179	1191	378	16	0	0	(12)
(13)		CDH80	1154	0	0	0	2	41	126	397	461	126	1	0	0	(13)
(14)	Wind	WSAvg	10.7	11.4	11.0	11.4	11.5	11.4	10.2	9.1	8.9	9.8	10.8	11.3	11.2	(14)
(15)	Precipitation	PrecAvg	14.8	0.7	0.6	0.7	1.0	2.0	2.5	2.1	1.4	1.5	0.7	0.6	0.8	(15)
(16)		PrecMax	20.1	1.5	1.3	2.0	3.0	5.1	6.7	5.2	4.7	3.9	3.7	1.6	2.1	(16)
(17)		PrecMin	8.8	0.1	0.1	0.0	0.0	0.0	0.6	0.8	0.1	0.1	0.0	0.0	0.2	(17)
(18)		PrecStd	3.1	0.4	0.4	0.4	0.8	1.2	1.5	1.3	1.1	1.1	0.8	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	45.9	61.8	75.6	84.8	89.9	93.6	95.4	89.0	76.0	61.9	46.9	(19)
(20)			MCWB	38.0	39.8	47.5	53.5	58.8	64.5	68.5	65.6	62.0	55.2	47.8	38.7	(20)
(21)		2%	DB	40.0	40.1	53.7	68.9	79.1	83.3	88.5	89.6	84.0	67.9	52.8	42.4	(21)
(22)			MCWB	35.5	35.6	43.7	50.7	56.9	63.7	68.1	64.8	60.3	52.1	42.9	36.4	(22)
(23)		5%	DB	36.9	36.4	48.1	63.9	74.5	79.4	84.3	85.2	78.1	61.8	47.8	38.7	(23)
(24)			MCWB	33.5	33.4	40.6	48.3	54.8	61.4	66.8	63.8	58.6	49.0	40.2	34.4	(24)
(25)		10%	DB	33.0	33.2	42.6	58.4	69.9	75.6	80.6	80.8	72.4	56.6	42.6	34.3	(25)
(26)			MCWB	30.4	31.1	37.6	45.6	52.6	59.8	65.1	62.9	56.1	46.6	37.1	31.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	39.8	48.3	55.7	62.0	70.6	75.2	71.4	65.7	57.3	48.3	39.7	(27)
(28)			MDCB	43.4	45.4	60.0	70.6	77.1	82.5	85.6	83.4	80.6	72.3	61.5	45.2	(28)
(29)		2%	WB	35.9	36.0	44.3	52.1	58.9	66.6	71.4	68.5	62.4	53.1	43.8	36.8	(29)
(30)			MDCB	39.6	39.2	52.3	66.3	74.2	77.9	81.5	81.5	76.8	65.4	51.8	41.6	(30)
(31)		5%	WB	33.7	33.5	40.9	49.3	56.7	63.8	68.8	66.3	59.8	50.1	40.5	34.4	(31)
(32)			MDCB	36.9	36.2	47.2	61.6	70.6	74.6	79.6	78.7	75.0	59.8	46.8	38.5	(32)
(33)		10%	WB	30.3	31.1	37.7	46.4	54.4	61.4	66.6	64.3	57.3	47.4	37.4	31.4	(33)
(34)			MDCB	32.9	33.2	42.3	56.9	67.0	72.0	77.7	77.8	70.8	55.8	41.8	34.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.1	18.3	18.6	23.4	25.2	22.1	24.4	26.4	25.5	21.5	17.9	16.9	(35)
(36)			MCDBR	19.2	19.7	25.5	33.0	32.4	28.8	31.3	35.0	35.9	31.4	25.3	19.8	(36)
(37)			MCWBR	15.9	15.8	16.5	17.7	15.6	13.3	14.6	14.6	16.6	17.8	16.4	15.0	(37)
(38)		5% WB	MCDBR	18.7	18.6	24.2	30.4	28.4	24.9	27.0	28.7	32.0	28.3	23.6	19.4	(38)
(39)			MCWBR	15.8	15.3	16.2	17.3	14.7	12.8	14.2	14.1	16.0	16.8	15.8	15.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.270	0.279	0.332	0.352	0.350	0.363	0.370	0.329	0.286	0.255	0.231	(40)	
(41)		taud	2.358	2.367	2.494	2.378	2.370	2.426	2.400	2.361	2.449	2.547	2.511	2.447	(41)	
(42)		Ebn at Noon	253	279	298	289	287	288	281	273	275	272	257	252	(42)	
(43)		Edh at Noon	20	26	27	35	36	35	35	35	29	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	404	740	1164	1530	1846	1933	2089	1658	1234	740	406	288	(44)	
(45)		RadStd	49	64	97	118	153	125	125	113	108	93	57	35	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.26	N/A	N/A	N/A	N/A	

Nomenclature: See separate page